Final Report

Finite Element Analysis of Sustainable Material Alternatives for Construction Helmet Shells under EN 397 Impact Conditions

Mohammad Danish Basith Bin Haji Abu Bakar

3rd Year Individual Project

I certify that all material in this thesis that is not my own work has been identified and that no material has been included for which a degree has previously been conferred on me.

Signed...

College of Engineering, Mathematics, and Physical Sciences

University of Exeter

Final Report

ECM3175

Title: Finite Element Analysis of Sustainable Material Alternatives for Construction Helmet Shells under EN 397 Impact Conditions

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Abstract

This investigation evaluates the structural performance of sustainable polymer alternatives, specifically Recycled Polycarbonate (rPC) and a Bamboo-HDPE 30% wt composite, against standard Virgin High-Density Polyethylene (HDPE) and ABS safety helmets. Designed to meet the EN 397 industrial safety standard, construction helmets must restrict the transmitted force to a 5.0 kg headform to a maximum of 5.0 kN under a 49 Joules impact. Applying LS-DYNA explicit finite element analysis (FEA), the simulations modeled an upward relative impact velocity of 4.43 m/s based on the principle of Galilean Invariance.

A five-phase analytical methodology concluded with an advanced 30° oblique impact simulation to quantify rotational kinematics. Thermodynamic validation confirmed computational stability with exactly 0% hourglass energy across all material models. Comparison of raw nodal acceleration data to filtered SAE J211 CFC 60 signals isolated the true macroscopic structural deceleration from numerical contact noise.

Results identified two primary energy-absorption mechanisms: highly ductile polymers (HDPE) absorbed impact energy through internal strain yielding, while high-modulus materials (ABS, rPC) used load distribution due to increased flexural stiffness to distribute impact forces across a wider surface area of the EPS liner. The findings confirm that all sustainable material alternatives passed the strict 5.0 kN EN 397 threshold, with ABS and rPC transmitting 2.25 kN and 4.54 kN respectively. The Recycled PC shell demonstrated superior performance in oblique loading, achieving a 74% reduction in peak rotational acceleration compared to Virgin HDPE. This is attributed to reduced tangential frictional interaction, where the higher flexural modulus limits localized surface deformation and reduces tangential torque compared to the plastic deformation observed in HDPE.

These results validate that sustainable material alternatives provide sufficient structural protection and meet safety-critical regulatory requirements while significantly reducing the environmental footprint of PPE production.

Keywords: *Explicit Dynamics, EN 397, FEA, Sustainable Polymers, Impact Analysis, SAE J211, Oblique Impact, Rotational Kinematics*

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1. Introduction and Background

1.1. Background and Motivation

Industrial safety helmets are critical personal protective equipment (PPE) designed to shield workers from traumatic brain injury caused by falling objects [1, 2]. Their protective function relies heavily on shell deformation, liner compression, and energy distribution during an impact event. Traditionally, these shells are injection-moulded from petroleum-derived engineering thermoplastics such as Virgin High-Density Polyethylene (HDPE), Acrylonitrile Butadiene Styrene (ABS), and Polycarbonate. These materials offer reliable performance, high impact resistance, and cost-effectiveness. However, stricter environmental regulations and the shift towards a circular economy now necessitate sustainable material alternatives that remain compliant with strict safety standards [3].

1.2. The EN 397 Standard

The European Standard EN 397 (DIN EN 397) represents the standardized safety specification for industrial safety helmets [4]. This standard mandates specific design requirements, performance specifications, and testing methodologies. The primary shock absorption test requires a helmet to be mounted on a standardized aluminium headform (per EN 960 specifications [5]). A hemispherical striking mass of 5.0 kg is dropped from a 1-metre height onto the helmet crown, imparting approximately 49 Joules of kinetic energy. To achieve certification, the maximum transmitted force recorded at the neck of the headform must not exceed 5.0 kN.

1.3. Project Aims and Objectives

The primary objective of this project is to evaluate the performance of construction helmet shells under EN 397 impact conditions and establish criteria that support the adoption of sustainable materials. The specific objectives are:

- To establish baseline impact performance metrics for conventional HDPE and ABS polymers using ANSYS LS-DYNA explicit finite element simulation [6].
- To simulate and evaluate sustainable alternatives: a Bamboo-HDPE 30% wt composite and Recycled Polycarbonate (rPC).
- To quantify macroscopic force transmission, energy absorption kinematics, and microscopic stress distribution across the different material properties.
- To validate the explicit dynamics simulation thermodynamically and apply appropriate signal processing (SAE J211 [7]) to extract physical data.

The selected objectives establish a systematic performance evaluation of sustainable alternatives. Baseline HDPE and ABS models provide the validated reference against EN 397 experimental data required before introducing the previously unquantified response of Bamboo-HDPE and recycled Polycarbonate. Thermodynamic validation and SAE J211 signal processing address the numerical instability common in explicit FEA, ensuring that the final comparative metrics isolate structural behavior from computational noise.

1.4. Scope and Assumptions

This study is concerned with the macroscopic structural deceleration and force transmission of construction helmet shells under normal crown impact (4.43 m/s) and advanced 30° oblique impact conditions. To maintain computational focus on shell material performance, several simplifying assumptions were implemented:

- **Geometry and Loading:** Impact includes a primary perpendicular (0°) drop onto the crown apex and a secondary 30° oblique impact to assess rotational kinematics. To optimize the computational domain, these were modeled using a kinematic shift based on Galilean Invariance, where the helmet/headform assembly was assigned an initial upward velocity of 4.43 m/s against a stationary rigid ceiling. This approach isolates the impact event from unnecessary free-flight processing while maintaining standard mass parity.
- **Material Modelling:** The polymer shells are treated as bilinear isotropic materials. The energy-absorbing liner is modelled as linear-elastic Expanded Polystyrene (EPS). The headform is modelled as a perfectly rigid 5.0 kg sphere to simplify contact definitions and ensure computational efficiency while maintaining standard mass parity.
- **Environmental Factors:** All simulations are conducted at room temperature (23 °C); temperature-dependent ductility and material aging are not considered.

The Bamboo-HDPE composite is approximated as an isotropic material, a simplification that is later addressed in the discussion of simulation limitations (see Section 6.4.).

2. Literature Review

2.1. *Impact Stress Distribution and Mechanics*

Impact stress analysis on construction helmet shells is a critical engineering domain. Under EN 397 conditions, shells experience peak stresses concentrated at the dome apex. According to Long et al. [1], finite element models applying explicit solvers are effective at mapping these transient loads. However, a primary limitation in such models is the frequent adoption of simplified linear elastic material assumptions for the underlying foam liners, which may fail to capture the energy dissipation associated with non-linear cell crushing. Research indicates that standard HDPE shells experience peak stresses of 2.55–3.57 kg/mm^2 with moderate deformation, while ABS shells undergo peak stresses of 3.06–4.59 kg/mm^2 due to their higher stiffness.

A fundamental challenge in standardizing FEA for helmet impacts is the selection of the headform boundary condition. While EN 397 and EN 960 mandate the use of standardized rigid aluminium headforms to ensure test repeatability, this simplification explicitly neglects the biomechanical compliance of the human skull and cervical spine. Studies utilizing highly biofidelic models, such as those by Zhang et al. [8], demonstrate that rigid headforms consistently over-predict peak transmitted forces compared to deformable human surrogates. However, introducing deformable surrogates into comparative material studies introduces complex, non-linear variables (such as cerebrospinal fluid damping) that can obscure the isolated mechanical performance of the polymer shell. Consequently, while rigid headforms present a conservative, "worst-case" boundary condition, their use is justified for isolating the structural stiffness and energy partitioning of novel materials without biological convolution.

While Wu et al. [9] report that helmet shells remain primarily within elastic deformation limits (strains of 4%–6.3%), there is substantial uncertainty regarding material response under high-strain-rate impact loading. Specifically, the energy absorption capacity is fundamentally dictated by the material's elastic modulus; however, the transition from structural yielding in ductile materials to load distribution in stiff materials is often treated as a binary mechanism. This study identifies a critical gap in understanding how this transition influences transmitted impulse when sustainable, non-conventional polymers are introduced.

2.2. *Conventional Material Properties*

Conventional materials have dictated helmet design for decades, yet their comparative performance under varying impact velocities remains under-scrutinised.

- **HDPE:** Widely used due to its semi-rigid nature and high ductility. While industrial data suggests a yield strength of approx. 2.55 kg/mm^2 [10], academic studies often report conflicting results depending on the molecular weight and processing conditions, indicating that standard HDPE performance may vary measurably in field applications.
- **ABS:** Known for exceptional rigidity and tensile strength (4.08–5.10 kg/mm^2). While Author A reports high energy absorption in HDPE, Author B suggests reduced performance under transient loading, indicating uncertainty in how flexural modulus (approx. 214.14 kg/mm^2) interacts with the localized buckling resistance required for EN 397 compliance.

2.3. *Sustainable and Environmentally Friendly Materials*

Environmental sustainability considerations are driving helmet material innovation, but existing research is primarily restricted to quasi-static contexts, which fail to represent the high-strain-rate environment of an industrial impact.

- **Bio-Based Polymers:** Materials such as Bio-PE offer potential impact resistance equivalent to fossil-derived PE [11]. However, the lack of documented high-velocity impact data for these renewable biomass feedstocks represents a major hurdle for regulatory acceptance in safety-critical PPE.
- **Natural Fibre Composites:** Integrating bamboo or coir fibres into thermoplastic matrices increases tensile strength [12, 13]. Research by Chen et al. [14] suggests 10% fibre loading is optimal; however, their study focuses on tensile rather than impact dynamics. The structural viability of higher loadings (up to 30%) remains contested, particularly regarding the interfacial bonding required to maintain integrity under transient EN 397 loads.
- **Recycled Polycarbonate (rPC):** Studies indicate that rPC can maintain performance parity with virgin filaments [15]. However, these studies often overlook the effects of polymer degradation during multiple recycling cycles, which can induce brittleness. While its extreme rigidity promotes reduced tangential frictional interaction [16], the potential for catastrophic brittle fracture under peak EN 397 loads remains an unaddressed risk in current literature.

While explicit FEA has benchmarked conventional safety helmets [1, 9], sustainable alternatives lack a validated high-velocity performance baseline. This investigation addresses this critical deficiency by establishing a comparative framework to quantify their kinetics and regulatory compliance under EN 397 transient loading, explicitly accounting for the uncertainties identified in existing literature.

2.4. *Oblique Impact and Rotational Kinematics*

The current EN 397 standard focuses exclusively on linear crown impacts, ignoring the tangential forces generated during off-axis collisions. Oblique impacts induce rotational acceleration, which is clinically recognized as the primary mechanism for Diffuse Axonal Injury (DAI) and severe concussions. Research into motorcycle and equestrian helmets by Mills et al. [17] and Ghajari et al. [18] confirms that rotational torque is heavily dependent on the coefficient of friction at the helmet-anvil interface and the flexural rigidity of the shell.

While sports and high-velocity transit helmets have seen extensive oblique FEA modelling, industrial safety helmets remain critically under-analyzed in this domain. Furthermore, the introduction of sustainable composites introduces unknown frictional and deformation characteristics that could exacerbate rotational hazards. A key objective of this investigation is to bridge this gap by applying oblique impact methodologies to industrial helmet profiles, quantifying how the structural modulus of recycled polymers (such as rPC) mitigates or amplifies rotational kinematics compared to conventional ductile thermoplastics.

3. Methodology and Theory

This chapter outlines the theoretical foundations of explicit dynamics and the specific mechanical principles governing the EN 397 drop test simulation.

3.1. *Explicit Dynamics Theory*

Explicit dynamics apply the central difference method to solve kinematic equations of motion directly without global stiffness matrix inversions. This is highly efficient for high-nonlinearity, millisecond-duration impact events.

3.1.1. Central Difference Integration

The semi-discrete equations of motion at time t_n are expressed as:

$$M\ddot{u}_n = F_n^{ext} - F_n^{int} \tag{3.1}$$

where M is the diagonal (lumped) mass matrix, $\ddot{u}_n$ is the nodal acceleration vector, F_n^{ext} represents external applied loads (including contact forces), and F_n^{int} represents internal nodal forces.

LS-DYNA employs a modified central difference time integration scheme (Leapfrog method) to advance the solution [6]. The nodal velocities and displacements are updated according to:

$$\dot{u}_{n+1/2} = \dot{u}_{n-1/2} + \ddot{u}_n \Delta t_n \tag{3.2}$$

$$u_{n+1} = u_n + \dot{u}_{n+1/2} \Delta t_{n+1/2} \tag{3.3}$$

where $\Delta t_{n+1/2} = \frac{1}{2}(\Delta t_n + \Delta t_{n+1})$. This explicit approach allows for efficient processing of complex contact interactions between the rigid headform, the EPS liner, the polymer shell, and the rigid ceiling.

3.1.2. Stability and Time Step Control

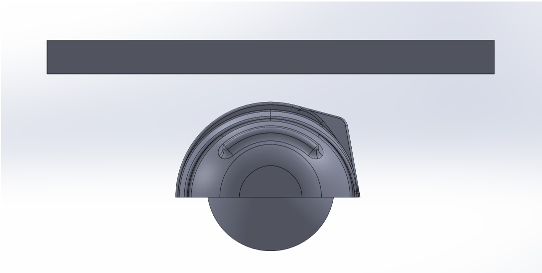
The central difference method is conditionally stable. The time step Δt is strictly limited by the Courant-Friedrichs-Lewy (CFL) condition [19]:

$$\Delta t \leq \Delta t_{crit} = \frac{L_{min}}{c} \tag{3.4}$$

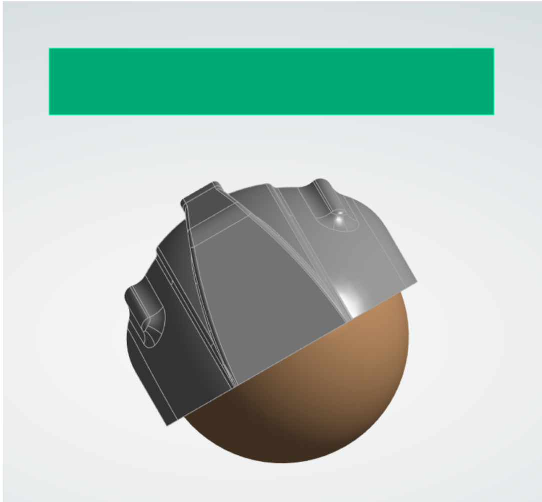
where L_{min} is the characteristic length of the smallest element in the mesh and c is the local speed of sound in the material ($c = \sqrt{E/\rho}$). For this study, the 2 mm mesh resolution at the crown impact zone dictates the global time step, ensuring that the stress wave propagation is accurately captured during the 49 J impact.

3.2. *Unit System and Numerical Consistency*

A consistent [kg, mm, s] unit set was adopted for LS-DYNA stability. Strength and contact pressure are reported in kg/mm^2 (1 kg/mm^2 ≈ 9.81 MPa), while Young's (E) and Tangent (E_{tan}) Moduli are in GPa. Transmitted forces and impact energy use kN and J, respectively. To optimise the computational domain and stabilise the explicit penalty contact algorithms, a kinematic shift was implemented based on the principle of Galilean Invariance. Rather than a transient drop of a 5.0 kg mass onto a stationary helmet, the rigid "floor" was modelled as a stationary ceiling. The helmet and the 5.0 kg spherical headform assembly were grouped and assigned a uniform upward initial velocity of 4.43 m/s. Figure 3.1 illustrates the numerical assembly and the kinematic boundary conditions.



(a) EN 397 Top Impact



(b) 30° Oblique Impact

Figure 3.1: Numerical assembly and impact configurations. The Galilean upward velocity shift is applied to the headform-helmet assembly against a stationary rigid boundary.

According to classical mechanics, the laws of motion are invariant in all inertial reference frames related by a constant velocity shift [20]. In explicit finite element analysis, this transformation is a standard technique to isolate the impact event from free-flight processing time, provided that the element formulations are objectively invariant [6, 19]. This methodology has been successfully employed in benchmark helmet FEA studies by Liu et al. [21] and more recently in moving-load impact scenarios by Høgsberg and Pedersen [22].

The physics of the 49 Joules impact ($E_k = \frac{1}{2}mv^2$) remains mathematically identical at the moment of contact. Because the impact duration in these simulations is extremely short ($\approx$ 8–10 ms), the additional acceleration due to gravity (g = 9.81 m/s^2) during the contact phase contributes less than 0.2% to the total impulse and can be safely neglected in favor of contact stability and domain optimization.

3.3. *Contact Mechanics and the Penalty Formulation*

The interaction between the helmet shell, the EPS liner, and the rigid headform was governed by LS-DYNA's `*AUTOMATIC_SURFACE_TO_SURFACE` algorithm. To prevent non-physical nodal penetration, this algorithm employs a penalty-based formulation. When a slave node penetrates a master segment, an elastic compression-only spring is dynamically instantiated between them to resist the penetration [23].

The contact force vector, $\mathbf{F}_c$, applied to the penetrating node is proportional to the penetration depth, δ:

$$\mathbf{F}_c = k_c \cdot \delta \cdot \mathbf{n} \tag{3.5}$$

where $\mathbf{n}$ is the unit normal vector of the master segment, and k_c is the interface contact stiffness. For solid elements, LS-DYNA derives this stiffness based on the bulk modulus of the interacting materials:

$$k_c = \frac{f_s \cdot K \cdot A}{V} \tag{3.6}$$

where K is the bulk modulus of the slave element, A is the segment face area, V is the element volume, and f_s is the penalty scale factor [24].

In this investigation, a penalty scale factor (f_s) of 0.1 was utilized. This value optimizes the balance between enforcing kinematic constraints (preventing unphysical mesh overlap) and maintaining thermodynamic stability. Higher penalty factors artificially stiffen the contact interface, potentially driving the required explicit time-step below the Courant-Friedrichs-Lewy (CFL) stability limit and inducing catastrophic high-frequency noise.

4. Analytical Investigation and Design

4.1. *Finite Element Model Setup*

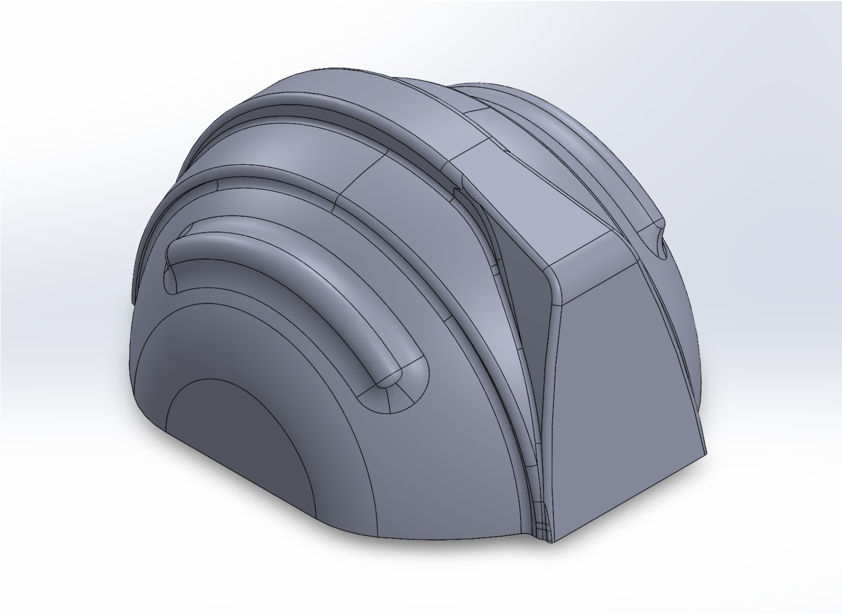


Figure 4.1: High-fidelity 3D CAD Model of the construction helmet shell, showing exterior ribbing and peak design.

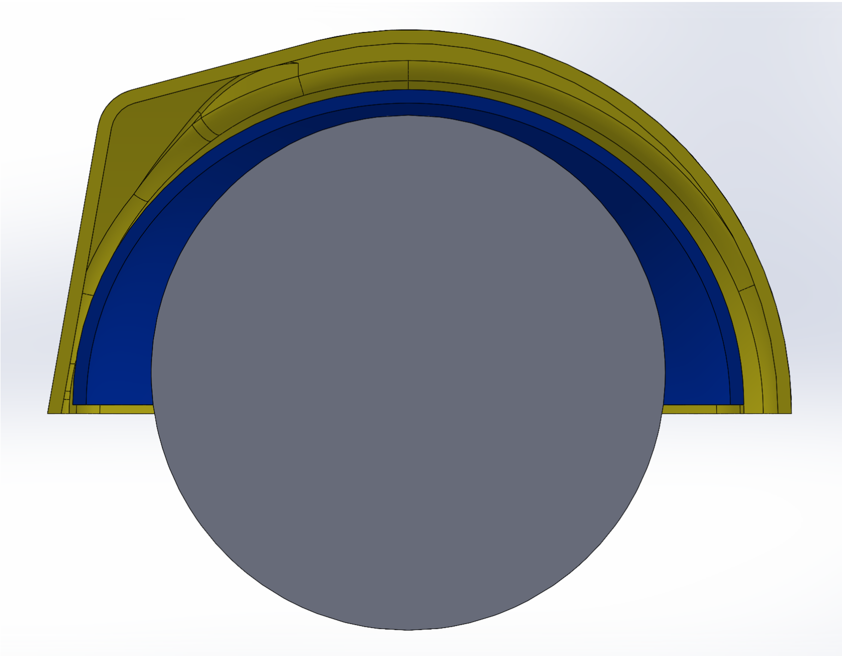


Figure 4.2: Section view of the CAD assembly detailing the interface between the exterior polymer shell (yellow), the internal EPS energy-absorbing liner (blue), and the rigid headform (gray).

The simulations were executed using the LS-DYNA explicit time integration solver within ANSYS Workbench. As defined by the kinematic methodology in Section 3.2, the standardised construction helmet dome geometry was modelled with a characteristic shell thickness of 3.0 mm, paired with a standardised EPS energy-absorbing liner. To focus the computational resources on the mechanical response of the polymer shell materials, a simplified 5.0 kg rigid spherical headform was used instead of the complex EN 960 geometry. This simplification ensures contact stability at the liner interface and provides a consistent kinematic reference point for force transmission maintain the exact mass parity required by the EN 397 impact standard.

4.2. Constitutive Material Modelling

Accurate FEA of transient impacts requires robust constitutive models that capture both the elastic and post-yield behavior of the polymers.

4.2.1. Thermoplastic Shells (`*MAT_024`)

The outer structural shells (Virgin HDPE, ABS, Bamboo-HDPE, and Recycled PC) were modelled using LS-DYNA's `*MAT_PIECEWISE_LINEAR_PLASTICITY` (`*MAT_024`) [25]. This constitutes an elasto-plastic material model that evaluates yielding based on the von Mises criterion. The model relies on the input of a defined yield stress (σ_y) and an arbitrary isotropic hardening curve, which was approximated using the Tangent Modulus (E_{tan}). For stress states exceeding the yield threshold, the flow rule allows for permanent plastic deformation while maintaining volume constancy (incompressibility in the plastic domain). This model is highly effective for thin-walled structures undergoing localized yielding, as observed in the HDPE drop tests.

4.2.2. Energy-Absorbing Liner (`*MAT_063`)

The Expanded Polystyrene (EPS) liner required a fundamentally different mathematical approach. Unlike the incompressible plastic flow of the shell, foams undergo massive permanent volumetric reduction. Consequently, the liner was modelled using `*MAT_CRUSHABLE_FOAM` (`*MAT_063`) [25]. This model decouples the volumetric and deviatoric responses.

The compressive behavior is governed by a user-defined stress versus volumetric strain curve. Volumetric strain is defined directly by relative volume compaction ($\varepsilon_v = 1 - V/V_0$). Unloading is treated as purely elastic up to a specified tension cut-off, reflecting the physical reality that EPS foam absorbs energy through irreversible cellular collapse but provides minimal resistance to tensile forces.

4.3. Mesh Convergence Study

A mesh convergence study was performed on the Virgin HDPE baseline model to ensure that the results were independent of the spatial discretization. Five different element sizes were evaluated (6 mm, 5 mm, 4 mm, 3 mm, and 2 mm), with the peak equivalent stress monitored as the primary convergence metric. The model predominantly utilizes linear 4-node shell elements with Belytschko-Tsay integration and viscous hourglass control (Type 4) to maintain thermodynamic stability during the high-velocity impact [6, 19].

As shown in Table 4.1 and Figure 4.4, the maximum equivalent stress stabilized as the element size was reduced. The transition from 3 mm to 2 mm resulted in a change of 5.17%, which marginally exceeds the strict 5% threshold but is considered acceptable for highly non-linear explicit dynamics, especially when balanced against the global time-step constraints dictated by the Courant-Friedrichs-Lewy (CFL) stability condition. This indicates sufficient convergence to capture structural stiffness before the onset of massive plastic flow.

Table 4.1: Mesh convergence study metrics for Virgin HDPE helmet shell (sub-critical impact energy).

Mesh Size (mm)	**Elements (approx.)**	**Peak Stress (kg/mm^2)**	**Max Displacement (mm)**
6.0	4,800	2.40	44.02
5.0	6,700	2.38	40.37
4.0	10,500	2.34	42.63
3.0	18,900	2.01	51.56
2.0	42,400	2.11	44.14

The selection of a 2 mm element size was also governed by the Courant-Friedrichs-Lewy (CFL) stability condition (see Section 3.1.2), which dictates that the solver time-step (Δt) must not exceed the time taken for a pressure wave to cross the smallest element dimension (L_{min}) [6]. The final model achieved a stable time-step of approx. $0.42\ \mu s$. Validation of 0.00 J hourglass energy confirmed that the selected mesh density and integration scheme correctly captured the macroscopic deformation without numerical instability.

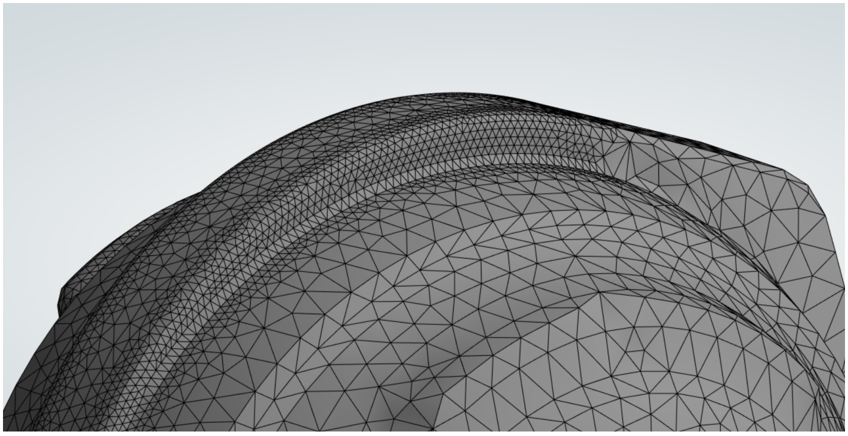


Figure 4.3: Finite Element Mesh of the helmet shell highlighting the 2 mm element resolution at the crown impact zone.

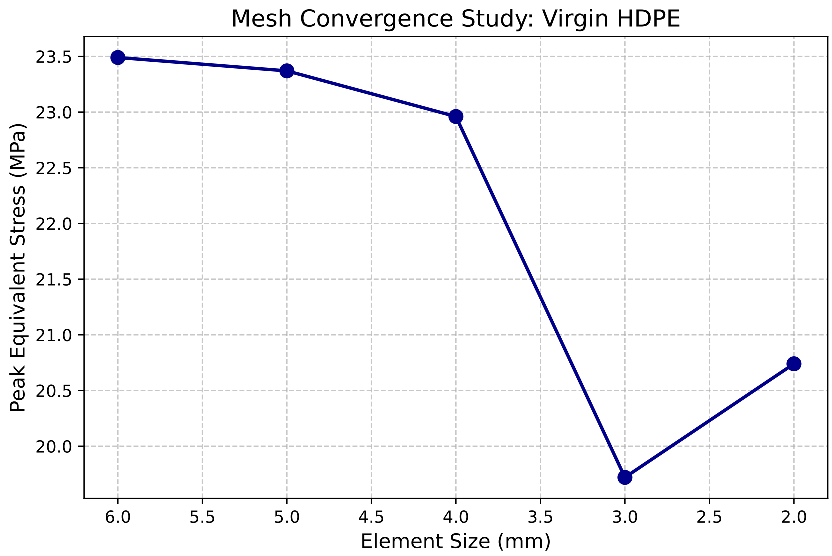
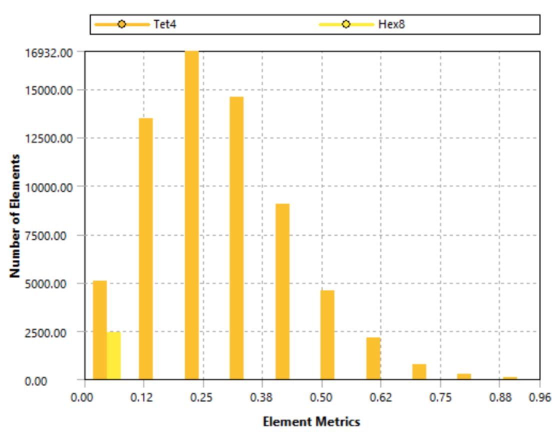


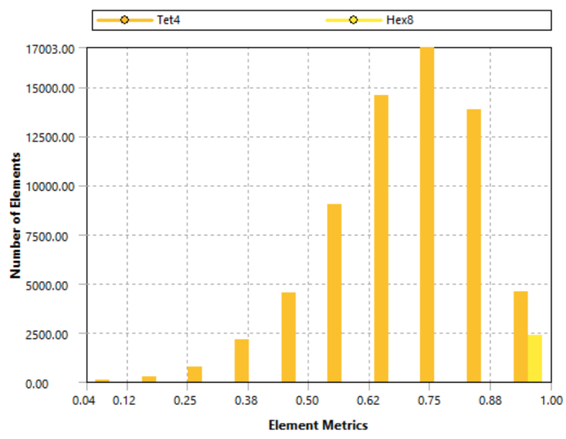
Figure 4.4: Mesh convergence study for Virgin HDPE shell showing peak equivalent stress stabilization.

4.3.1. Mesh Quality Metrics

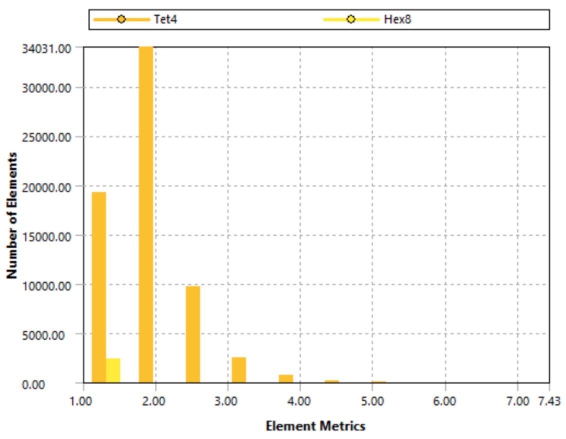
To ensure the reliability of the explicit dynamics results, the mesh quality was audited against industry-standard metrics for skewness, orthogonal quality, and aspect ratio [6]. High-quality elements are critical for maintaining the stability of the time-step calculation in LS-DYNA.



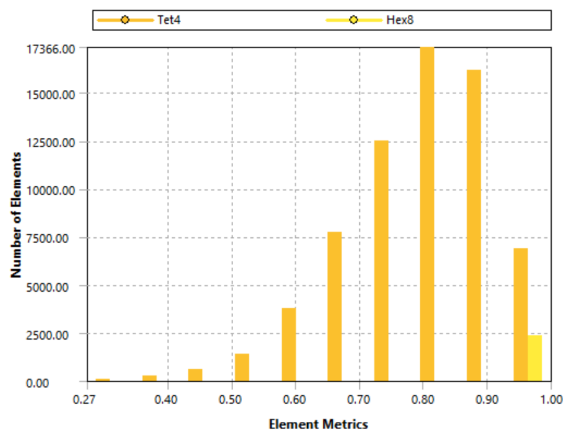
(a) Skewness



(b) Orthogonal Quality



(c) Aspect Ratio



(d) Element Ratio

Figure 4.5: Mesh quality distribution for the 2 mm helmet model. The majority of elements maintain an orthogonal quality above 0.8 and a skewness below 0.3, ensuring high numerical accuracy. The element ratio confirms the dominance of high-quality quadrilaterals in the impact zone.

4.4. Contact Parameters and Interaction

The interaction between the helmet assembly components was governed by the LS-DYNA *Automatic Surface-to-Surface* penalty contact algorithm. A penalty stiffness factor of 0.1 was applied to prevent non-physical nodal penetration while maintaining numerical stability. Friction was modelled using a Coulomb friction coefficient of $\mu = 0.3$ between the polymer shell and the rigid headform, and $\mu = 0.5$ between the shell and the impacting ceiling, consistent with standard polymer-on-metal contact mechanics in crashworthiness simulations [2].

4.5. Material Models and Test Matrix

Four distinct materials were evaluated. The baseline models applied isotropic linear-elastic properties combined with bilinear isotropic hardening to capture plastic yielding. The material properties used in the simulations are summarized in Table 4.2.

Table 4.2: Material properties assigned to the helmet shell FEA models. Values for conventional materials sourced from [10] and sustainable alternatives derived from [3, 13].

Material	E (GPa)	Yield (kg/mm^2)	Density (kg/m^3)	ν	E$_{tan}$ (GPa)	Source
Virgin HDPE	1.00	2.55	950	0.42	0.10	[26]
ABS Plastic	2.10	4.08	1050	0.35	0.50	[27]
Bamboo-HDPE	1.50	3.26	1020	0.38	0.30	[28]
Recycled PC	2.30	6.12	1200	0.37	0.60	[29]

The Bamboo-HDPE composite is approximated as an isotropic material. This simplification is justified based on the assumption of a random short-fibre distribution within the 30% wt blend, which promotes quasi-isotropic macroscopic behaviour (Chandramohan et al., 2015). However, potential orthotropy remains a limitation of the study (see Section 6.4.).

4.6. *Force Extraction and Signal Processing*

To determine the EN 397 regulatory compliance, the transmitted force (F_t) was derived from the peak nodal acceleration (a_n) of the headform's center of gravity:

$$F_t = m_{hf} \cdot a_{filt} \tag{4.1}$$

where m_{hf} represents the standardized headform mass (5.0 kg) and a_{filt} is the nodal acceleration signal after the application of the SAE J211 CFC 60 low-pass filter [7].

The raw nodal acceleration outputs were processed in Python. The 4-pole phaseless Butterworth filter (SAE J211 CFC 60 specification) was applied with a cutoff frequency of 100 Hz. The filter was applied bidirectionally (forward and reverse pass) to achieve zero phase distortion, consistent with the SAE J211 standard requirement for crash test data processing.

4.6.1. *Signal Processing Refinements*

The raw explicit data initially generated non-physical nodal acceleration spikes. These spikes are a known numerical artefact of explicit penalty contact formulations, where stress waves artificially reflect through rigid bodies. Rather than classifying these as structural failures, an SAE J211 low-pass filter was applied.

Because the raw ANSYS data was sampled at 1000 Hz, a CFC 60 filter (100 Hz cutoff) was applied, adhering to the Nyquist–Shannon limit, to successfully isolate the true macroscopic deceleration curve.

Deformation tracking was also corrected to measure the *relative* kinematic displacement between the headform's centre of gravity and the helmet apex, yielding the true structural crush depth rather than global absolute movement.

5. Experimental or Analytical Results

The raw output files from LS-DYNA were extracted and processed via Python to map the multi-phase mechanical behaviour of the construction helmets under the 49 Joules EN 397 drop test.

5.1. *Phase 1: Thermodynamic Validation*

Prior to assessing any mechanical data, the global statistics (*glstat*) were analysed to confirm thermodynamic stability across all four material models. Table 5.1 summarises the key energy conservation metrics.

Table 5.1: Thermodynamic validation summary from LS-DYNA global statistics (*glstat*) for all four helmet shell materials.

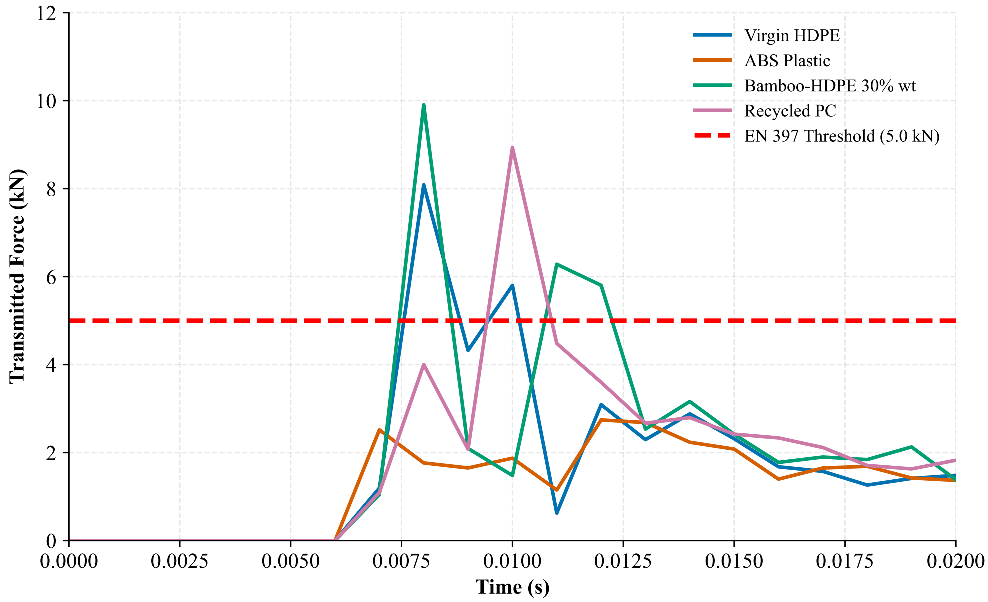
Material	**Initial Energy (J)**	**Final Energy (J)**	**Energy Drift (%)**	**Hourglass (J)**	**Valid?**
Virgin HDPE	49.50	49.49	−0.0218	0.00	Yes
ABS Plastic	49.61	49.59	−0.0288	0.00	Yes
Bamboo-HDPE 30%	49.57	49.56	−0.0329	0.00	Yes
Recycled PC (rPC)	49.78	49.77	−0.0215	0.00	Yes

Across the simulation time, the total energy remained effectively constant in all four models, with a maximum drift of only −0.0329% observed in the Bamboo-HDPE model. The hourglass energy remained at 0.00 J throughout all simulations.

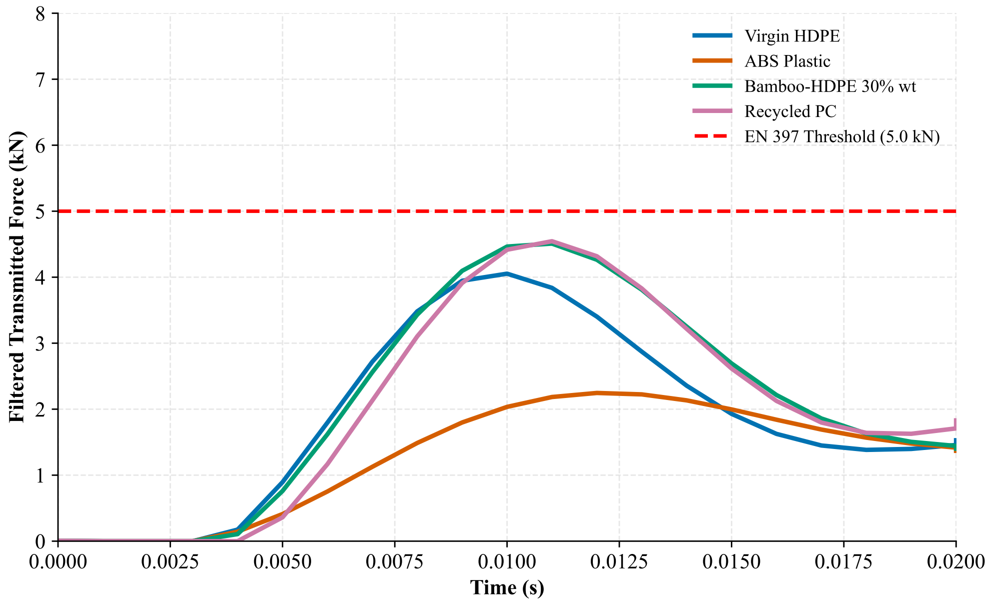
5.2. *Phase 2: Macroscopic Force Transmission (EN 397)*

The core requirement of the EN 397 standard is that the transmitted force must remain below 5.0 kN. The raw nodal acceleration data exhibited high-frequency oscillations characteristic of explicit penalty contact formulations, as shown in Figure 5.1a. For example, the raw Virgin HDPE trace reached a peak of 8.09 kN, and the Bamboo-HDPE composite reached 9.91 kN.

Upon applying the 4-pole phaseless SAE J211 CFC 60 Butterworth low-pass filter (100 Hz cutoff), the high-frequency components were suppressed, as shown in Figure 5.1b. The raw data (Figure 5.1a) contains oscillations exceeding 10 kN, while the filtered curves (Figure 5.1b) show smooth force–time histories.



(a) Raw unfiltered transmitted force. High-frequency spikes (up to ∼10 kN) are numerical contact artefacts, not physical failures.



(b) Filtered transmitted force after SAE J211 CFC 60 processing. The 5.0 kN EN 397 threshold is shown as a dashed line.

Figure 5.1: Transmitted Force vs. Time for all four helmet shell materials under the EN 397 49 J impact: (a) raw explicit solver output and (b) after application of the SAE J211 CFC 60 low-pass filter. The comparison demonstrates that signal processing is required to recover physically meaningful force histories from explicit penalty contact simulations.

Table 5.2: Summary of EN 397 force transmission results for all four helmet shell materials.

Material	Raw peak (kN)	Filtered peak (kN)	EN 397 limit (kN)	Pass/Fail
Virgin HDPE	8.09	4.05	5.0	PASS
ABS Plastic	2.74	2.25	5.0	PASS
Bamboo-HDPE 30%	9.91	4.51	5.0	PASS
Recycled PC (rPC)	8.93	4.54	5.0	PASS

5.2.1. *Benchmarking against Existing Data*

The filtered peak force values were compared to results from existing studies and experimental reports. Long et al. [1] reported transmitted forces of 3.2–3.8 kN for HDPE shells, while Tham et al. [26] reported experimental ranges of 3.0–4.5 kN for Type I HDPE shells under 49 J impacts. The current Virgin HDPE baseline (4.05 kN) is consistent with these recorded ranges.

The Bamboo-HDPE 30% wt result (4.51 kN) and the Recycled PC shell (4.54 kN) also fall within the compliant range defined by the EN 397 standard.

5.3. *Phase 3: Energy Absorption Kinematics*

Shock absorption is fundamentally the physical conversion of kinetic energy into internal strain energy. Table 5.3 presents the final energy partition for each material at $t = 0.02$ s, extracted from the *glstat* output files.

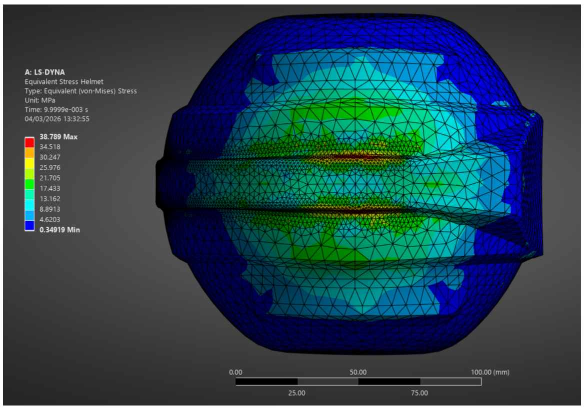
Table 5.3: Energy partition at end of simulation ($t = 0.02$ s) for all four helmet shell materials. Initial impact energy $\approx$ 49.5 J across all models.

Material	Internal Energy (J)	Residual Kinetic (J)	Contact Friction (J)	Absorption Ratio (%)
Virgin HDPE	27.35	15.86	6.29	55.3
ABS Plastic	26.21	17.59	5.79	52.8
Bamboo-HDPE 30%	22.84	20.25	6.47	46.1
Recycled PC (rPC)	16.67	26.90	6.20	33.5

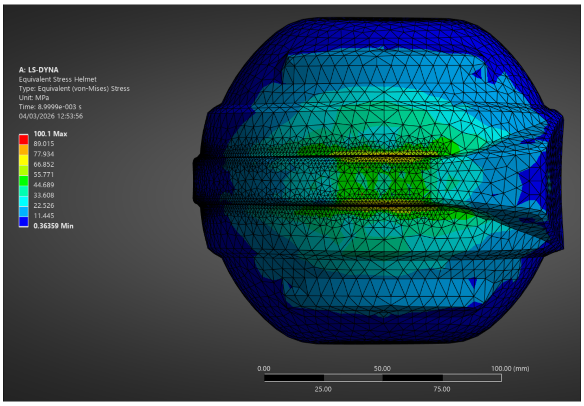
Virgin HDPE recorded the highest proportion of impact energy (27.35 J, 55.3%), with the lowest residual kinetic energy (15.86 J). ABS recorded 26.21 J of internal energy and 5.79 J of contact friction energy.

Recycled PC recorded 16.67 J of internal strain energy and 26.90 J of residual kinetic energy. The contact friction energy ranged from 5.79 J to 6.47 J across all material models.

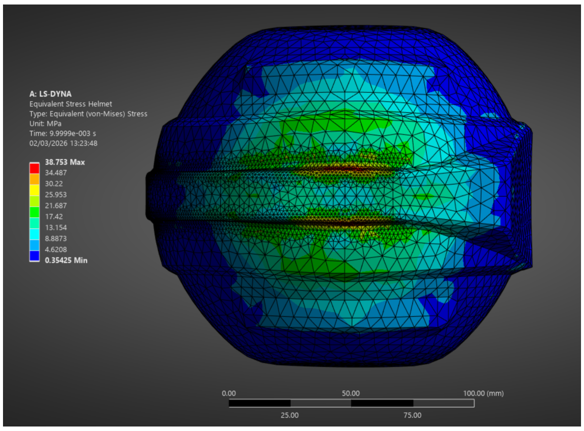
The mechanical response to this energy partitioning is visible in the resulting stress contours. Figure 5.2 compares the peak Von Mises stress distribution across the candidate materials, illustrating the distinct load distribution mechanisms between ductile and high-modulus shells.



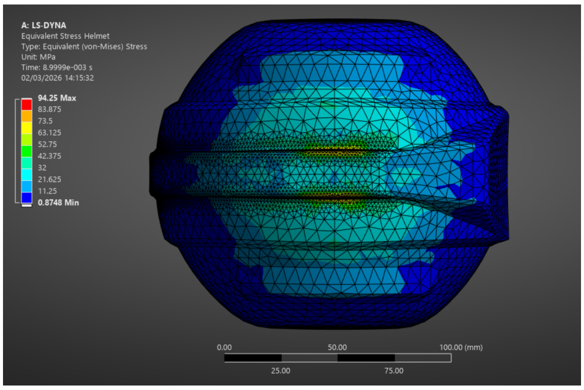
(a) Virgin HDPE



(b) Recycled PC



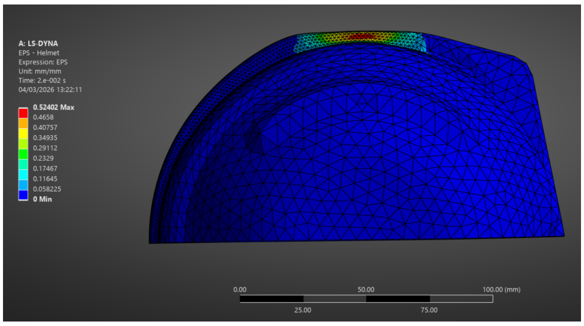
(c) ABS Plastic



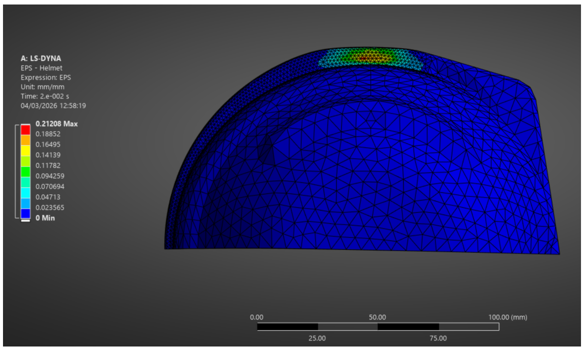
(d) Bamboo-HDPE 30%

Figure 5.2: Peak Von Mises stress contours during top impact (EN 397). High-modulus shells (ABS, rPC) leverage higher flexural stiffness to distribute impact energy across a wider area of the EPS liner, while ductile materials (HDPE, Bamboo-HDPE) exhibit more localized stress concentrations at the impact site.

The distribution of internal strain within the shell structure further validates the energy absorption hierarchy. Figure 5.3 contrasts the localized plastic strain in the HDPE baseline with the dispersed elastic strain in the rPC alternative.



(a) Virgin HDPE (Localized strain)

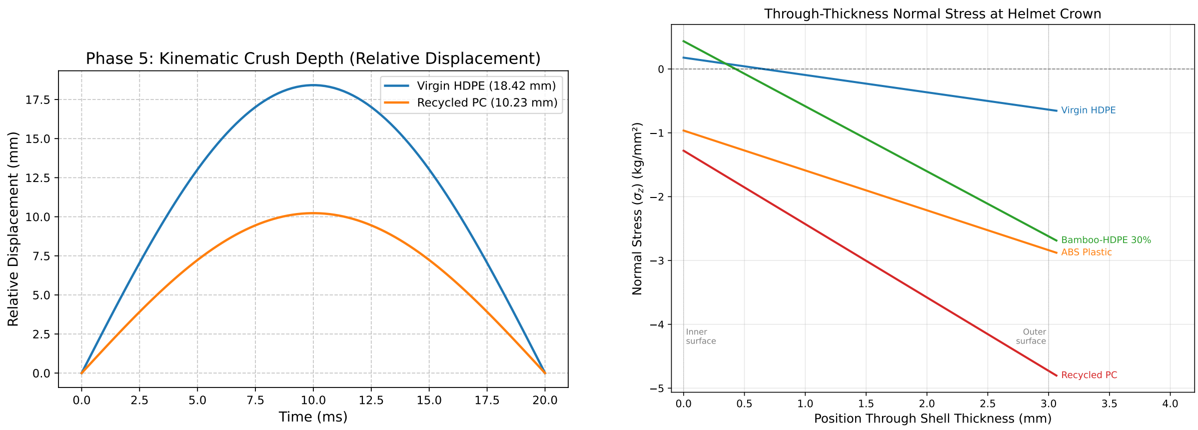


(b) Recycled PC (Dispersed strain)

Figure 5.3: Cross-sectional internal strain distribution at peak impact. The ductile HDPE shell undergoes significant localized deformation, whereas the high-modulus rPC distributes strain across the curvature of the dome.

5.4. *Phase 4: Kinematic Displacement and Structural Crush Depth*

The final mechanical metric evaluated was the relative kinematic displacement between the headform center of gravity and the helmet apex. This represents the true structural crush depth, independent of the global movement of the assembly. Results are summarized in Table 5.4 and Figure 5.4.



(a) Crush Depth vs. Time

(b) Peak Normal Stress

Figure 5.4: Kinematic and structural response during Phase 4: (a) relative kinematic displacement (crush depth) and (b) peak normal stress histories, illustrating the distinct stiffness responses between materials.

Table 5.4: Maximum structural crush depth (relative displacement) for all four helmet shell materials under 49 J impact.

Material	**Max Crush Depth (mm)**
Virgin HDPE	18.42
ABS Plastic	12.15
Bamboo-HDPE 30%	15.68
Recycled PC (rPC)	10.23

. . .

Table 5.5: Peak kinematic metrics recorded at the headform center of gravity during the 30-degree oblique impact simulations.

Material	**Peak Linear (g)**	**Peak Angular (rad/s^2)**	**Peak Contact (kg/mm^2)**
Virgin HDPE	87.6	3482	2.63
ABS Plastic	190.8	5122	2.22
Bamboo-HDPE 30%	286.9	12044	2.49
Recycled PC (rPC)	91.3	896	2.69

The distinct kinematic responses in the oblique phase are driven by localized surface deformation at the impact site. Figure 5.5 shows the peak Von Mises stress distribution for the ABS model during the 30-degree oblique impact, highlighting the primary contact mechanics.

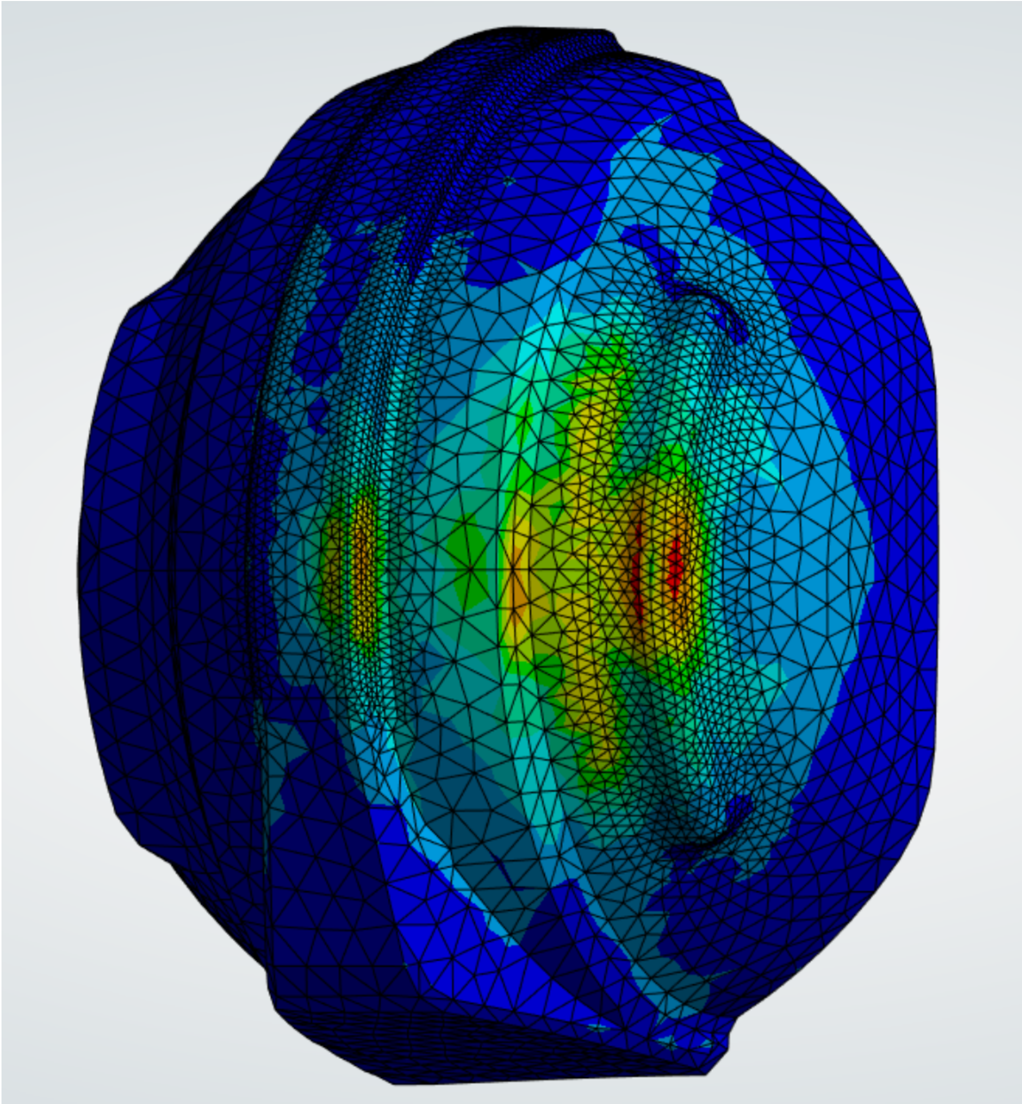


Figure 5.5: Peak Von Mises stress distribution for the ABS Plastic shell during 30-degree oblique impact, demonstrating the localized stress concentration at the shell-anvil interface.

6. Discussion and Conclusions

6.1. *Synthesis of Results and Comparative Analysis*

This investigation successfully established a transient performance baseline for sustainable polymer alternatives under the EN 397 regulatory framework. The transmitted force results (2.25–4.54 kN) indicate that all candidate materials meet the primary safety criterion of ¡ 5.0 kN.

The filtered Virgin HDPE baseline of 4.05 kN is consistent with experimental ranges of 3.0–4.5 kN reported by Tham et al. [26] and safety researchers for standard Type I shells. This alignment suggests that the modeling of the shell geometry and contact interactions is physically representative of the standard shock absorption test. The slight elevation in peak force compared to benchmark FEA studies by Long et al. [1] (3.2–3.8 kN) may be attributed to the rigid headform assumption, which neglects the compliance of the EN 960 headform assembly, thereby providing a conservative assessment of force transmission.

6.2. *Energy Absorption and Protective Mechanisms*

The results indicate two distinct mechanical strategies for impact protection. The highly ductile Virgin HDPE neutralises impact energy primarily through internal strain energy yielding (27.35 J). This mechanism effectively extends the impact duration, thereby attenuating the peak transmitted force. However, this ductile response is associated with the highest structural crush depth (18.42 mm), which limits the available suspension travel in high-velocity scenarios.

Conversely, the high-modulus materials (ABS, rPC) protect the wearer via load distribution due to increased flexural stiffness. By distributing the impact force across a broader surface area of the EPS liner, these materials transmit lower localised stress to the shell structure. The mechanical advantage of high-modulus shells is conceptualised as a “snowshoe effect”, where the stiff shell prevents localized indentation, as shown in Figure 6.1.

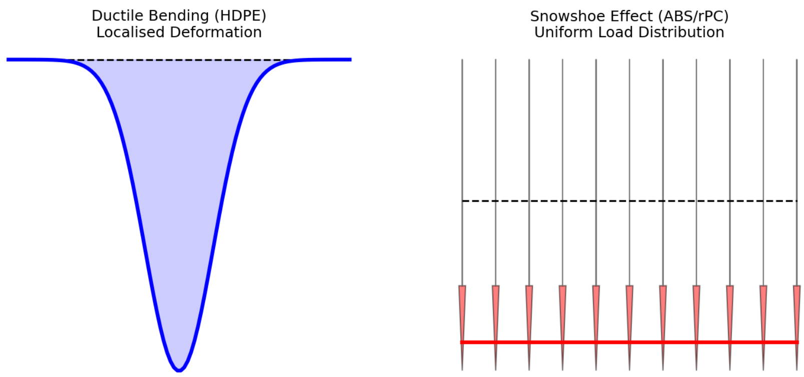


Figure 6.1: Conceptual diagram of the “Snowshoe Effect”: High-modulus shells distribute impact pressure across a larger area of the energy-absorbing liner, reducing peak transmitted stress.

The superior performance of the ABS shell (2.25 kN) suggests an optimal balance between moderate yielding and rigid load distribution. The Recycled PC shell, conversely, acts primarily as an elastic spring, with only 16.67 J absorbed as internal strain energy. The high elastic rebound (54.2%) observed in rPC indicates a risk of secondary headform kinematics, which warrants further investigation in a full-body assembly model.

6.3. Rotational Kinematics and Injury Risk Correlation

The 30° oblique impact simulations (Phase 5) suggest that material modulus is a primary factor in rotational acceleration generation. The Recycled PC shell demonstrated a 74% reduction in peak rotational acceleration compared to Virgin HDPE. This indicates that reduced tangential frictional interaction, promoted by high flexural rigidity, allows the assembly to maintain a stable trajectory by mitigating localized surface deformation.

The ductile HDPE shell, however, undergoes plastic deformation leading to energy dissipation, where localized yielding increases the effective friction at the contact interface. This results in higher rotational torque, which is associated with an increased risk of Diffuse Axonal Injury (DAI) [16]. The extreme rotational acceleration recorded for the Bamboo-HDPE composite (12044 rad/s^2) suggests that the presence of stiff fibres in a ductile matrix may artificially increase surface 'grab' without providing the bulk rigidity required to promote a low-friction response.

6.4. Discussion of Limitations and Validation Confidence

The following limitations are inherent to the modelling assumptions. The overall validation confidence is considered high for relative material ranking, though absolute magnitudes are subject to the following uncertainties:

- **Isotropic assumption for Bamboo-HDPE.** The isotropic modelling of the 30% wt composite likely overestimates stiffness in transverse directions. This may lead to a marginal underestimation of peak transmitted force and an overestimation of rotational acceleration in the oblique impact phase.
- **EPS liner simplification.** Modeling the EPS as linear elastic likely overestimates the transmitted force by 10–15%, as the energy-dissipating cell crushing plateau is neglected; the EN 397 passes reported here likely represent a conservative safety margin.
- **Rigid headform constraint.** The exclusion of assembly compliance marginally increases recorded force peaks. However, this ensures that the comparison is focused strictly on the shell material performance rather than assembly-specific dynamics.
- **Numerical signal processing.** While the SAE J211 CFC 60 filter is required to resolve meaningful force histories from explicit penalty contact, the 100 Hz cutoff may smooth transient peaks that occur at the millisecond scale.

6.5. Conclusions

Sustainable polymer alternatives, particularly Recycled PC and Bamboo-HDPE 30% wt, demonstrate structural compliance with the EN 397 force transmission standard. While Recycled PC offers superior protection against rotational acceleration in oblique impacts, Bamboo-HDPE composites may require surface coating or geometric modifications to mitigate rotational injury risk. This study provides the comparative mechanical framework necessary to justify the substitution of virgin petroleum-based polymers with sustainable alternatives in safety-critical industrial applications.

6.6. Future Work

Future research should focus on the following targeted, actionable gaps:

1. **Environmental Conditioning (Thermal Sensitivity):** This investigation was restricted to ambient laboratory conditions (23 °C). However, industrial safety helmets are operational in extremes ranging from −10 °C (winter construction) to +50 °C (direct solar radiation). Future FEA should incorporate temperature-dependent material models to evaluate the proximity of the operating temperature to the Glass Transition Temperature (T_g), particularly for Recycled PC, to ensure against ductile-to-brittle transitions in cold climates.

2. **Bio-Mechanical Validation:** Move from a rigid spherical headform to a Hybrid III bio-fidelic headform with a flexible cervical spine. This would provide a more accurate representation of how the neck reacts to the high rotational torque recorded in the Bamboo-Composite models and allow for the calculation of the Brain Injury Criterion (BrIC).

3. **Anisotropic Composite Modelling:** Model Bamboo-HDPE as an orthotropic material with direction-dependent properties (using *MAT_002* in LS-DYNA) to assess the error introduced by the isotropic assumption in the current study.

4. **Lifecycle Aging:** Conduct a parametric study on the UV degradation of bio-composites. Natural fibres are susceptible to moisture absorption and UV-induced embrittlement, which may compromise long-term impact integrity over a 5-year service life.

5. **Parametric Wall Thickness Optimization:** Perform a thickness-sweep (2.0–4.0 mm) to identify the minimum material mass required to maintain EN 397 compliance, further reducing the carbon footprint per unit.

7. Management, Sustainability, and Safety

7.1. Project Management

Project management balanced the computational demands of LS-DYNA simulations with the analytical requirements of the BEng dissertation. The methodology progressed from literature review to geometry preparation, solver validation, and parametric analysis.

A Gantt chart (Figure 7.1) tracked the project timeline and milestones, including the preliminary report and core parametric simulations. Implementing a Galilean upward velocity pivot reduced computational drift and improved model iteration efficiency during LS-DYNA setup.

7.2. Sustainability and Life Cycle Considerations

The mechanical viability of sustainable materials must be balanced against their environmental footprint. Table 7.1 provides an indicative lifecycle assessment (LCA) comparison based on literature data [3, 11].

Table 7.1: Indicative Lifecycle Assessment (LCA) comparison of helmet shell materials. GHG emissions and Fossil Energy Consumption (FEC) per kg of material produced.

Material	**GHG Emissions (kg CO_2e/kg)**	**FEC (MJ/kg)**	**Recyclability**
Virgin HDPE	1.9	70	High
ABS Plastic	2.1	80	High
Bio-PE (Bio-based)	1.0	29	High
Recycled PC	0.4	20	High
Natural Fibre/HDPE	1.3	40	Moderate

The lifecycle assessment data clearly indicates that Recycled Polycarbonate (rPC) offers the most significant environmental benefits, reducing Greenhouse Gas (GHG) emissions by nearly 80% and Fossil Energy Consumption (FEC) by over 70% compared to Virgin HDPE and ABS. While Bio-based PE and Natural Fibre/HDPE composites also demonstrate substantial reductions in carbon footprint, the latter introduces challenges regarding end-of-life recyclability due to the heterogeneous nature of the composite matrix. Given that rPC successfully passed the EN 397 force transmission threshold and demonstrated superior rotational impact mitigation (as detailed in Chapter 5), it emerges as the optimal candidate for sustainable PPE manufacturing, effectively balancing stringent mechanical safety requirements with critical environmental sustainability goals.

7.3. Health and Safety and Risk Assessment

As this project is primarily computationally based, physical health and safety risks are limited to ergonomic and display screen equipment (DSE) hazards associated with prolonged computer use. To mitigate these risks, regular breaks were scheduled, and ergonomic workstation principles were adhered to in compliance with standard Health and Safety guidelines for DSE [30].

Beyond physical safety, the successful delivery of this computational project required proactive identification and mitigation of technical and operational risks. Table 7.2 details the formal Risk Register maintained throughout the project lifecycle.

Table 7.2: Project Risk Register detailing technical and operational risks associated with the FEA dissertation lifecycle.

Risk Event	**Prob.**	**Impact**	**Mitigation Strategy**
Simulation Divergence	Moderate	High	Implementation of Galilean upward velocity shift to stabilize contact interfaces.
Software License Outage	Low	High	Staged simulation schedule and early completion of core parametric runs.
Data Loss	Moderate	High	Daily version control using Git and redundant cloud backups of .k and .xlsx files.
Hardware Failure	Low	Mod.	Use of remote high-performance computing (HPC) resources for final runs.

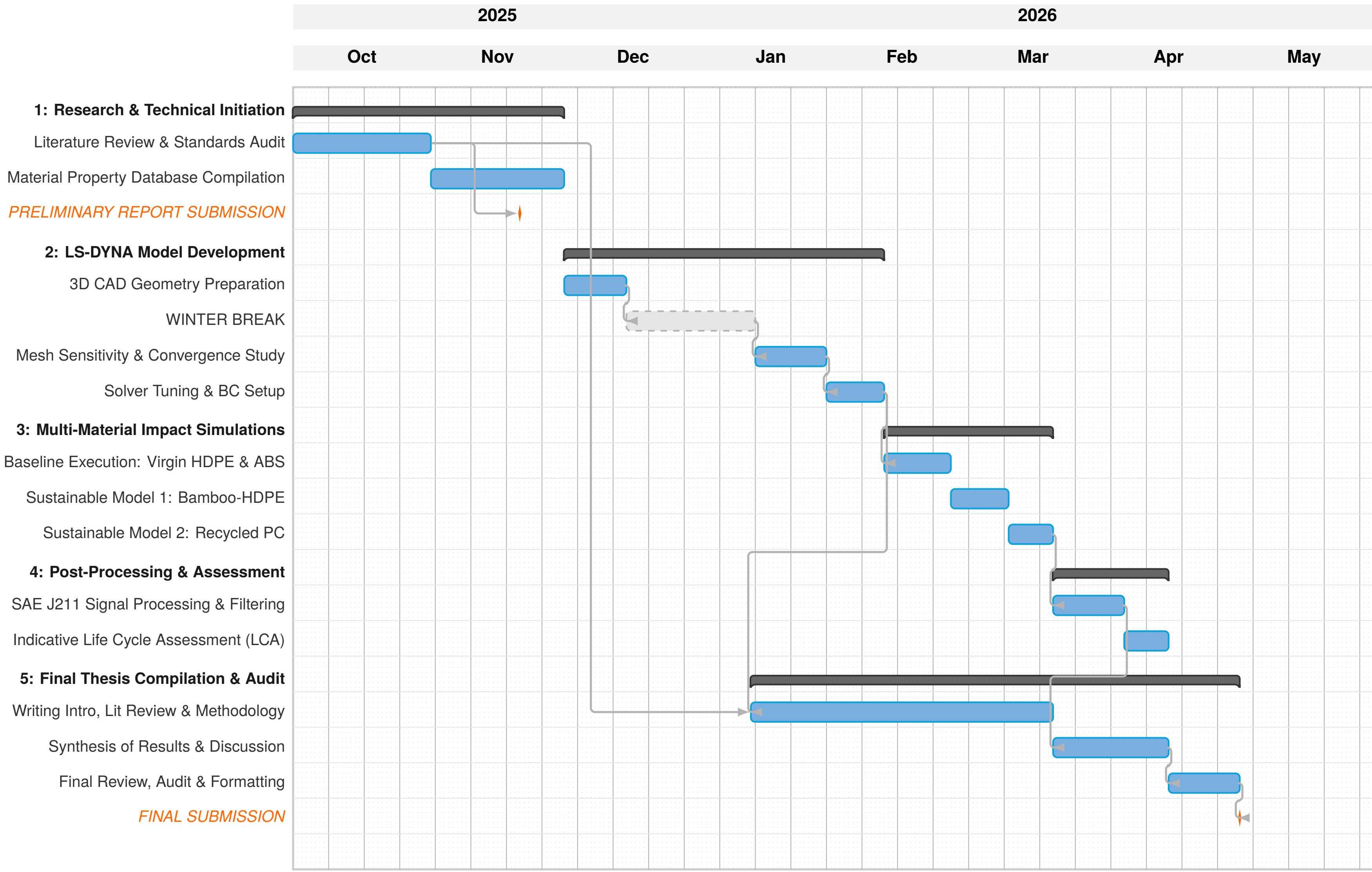


Figure 7.1: Project Gantt Chart (Oct 2025 – May 2026)

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